

SOP-SK-05 — Spike Execution

Kode	SOP-SK-05
Pemilik	Senku (R&D Engineer)
Revisi	1.0 · Berlaku 2026-05-29 · Reviu per kuartal
Referensi	Time-boxed spike, hypothesis quality test, tools/poc-report-template.md
COBIT	APO04 (Managed Innovation)

1. Tujuan

Memvalidasi **kelayakan teknis** lewat prototype **time-boxed (≤ 3 hari, throwaway)** — confirm/refute hipotesis dengan bukti, bukan asumsi — tanpa kebablasan ngoding production.

2. Ruang Lingkup

- **Berlaku:** pertanyaan teknis yang butuh kode untuk dijawab (integrasi novel, perf claim, fit-stack uncertain).
- **Tidak berlaku:** pertanyaan yang bisa dijawab literature-scan (well-documented → SOP-SK-01); shipping fitur (itu Killua/Saitama).

3. Definisi & Istilah

- **Spike** — eksperimen time-boxed buat validate satu hipotesis teknis.
- **Throwaway code** — kode prototype yang tujuannya validate, **bukan di-ship** (dirty OK).
- **Stop condition** — batas yang menghentikan spike: time-box habis ATAU bukti cukup.
- **Time-box** — alokasi waktu maksimum: tight 2–4 jam / medium 1 hari / long 3 hari (hard cap).

4. Referensi

Tata mandate (prototype ≤ 3 hari, throwaway OK, ship bukan tugas Senku), hypothesis quality test (PLAYBOOK §5.5), reversibility (Type-1/Type-2).

5. Tanggung Jawab (RACI)

Aktivitas	R	A	C	I
Eksekusi spike	Senku	Senku	@kakashi/@saitama (kalau perlu input teknis)	@lelouch
Hand-off ke production (kalau keep)	@killua/ @saitama	@kakashi	Senku	Tata
Report hipotesis gugur	Senku	Senku	—	@lelouch, Tata

6. Prosedur

6.1 Setup

- 6.1.1 Tulis **hipotesis falsifiable + metric + stop condition**. Lewatkan hypothesis quality test (Specific/Measurable/Falsifiable/Time-boxed/Relevant).
- 6.1.2 Pilih **time-box** (tight/medium/long) sesuai kompleksitas (PLAYBOOK §5.2).
- 6.1.3 Decision: **build POC vs research-only** (§5.3) — kalau well-documented, gak usah spike.
- 6.1.4 Siapkan workspace throwaway: `team/senku/spikes/<date>-<topic>/`. Daemon: `alfred.implement(task, project_dir, acceptance="prototype validates X within Y min")` async.

6.2 Eksekusi

- 6.2.1 Build prototype **minimal** — cukup buat jawab hipotesis, jangan polish.
- 6.2.2 Kumpulkan **bukti** (screenshot, log, angka, benchmark).
- 6.2.3 **Decision point — time-box habis?** → **STOP** walau belum kelar. Hasil partial = tetap finding.
- 6.2.4 **Exit kalau > 3 hari** → **STOP + argue ke @lelouch** sebelum lanjut (jangan grinding diam).

6.3 Conclude

- 6.3.1 Tulis **spike report** (poc-report-template): hipotesis, method, finding (worked/didn't), evidence, conclusion (confirmed/refuted/inconclusive), confidence + rationale.
- 6.3.2 **Decision point**: hipotesis gugur? → **report langsung** ke @lelouch + Tata (anti-hide), pivot. Confirmed + likely keep? → hand-off @killua/@saitama untuk rebuild proper (throwaway gak di-ship).
- 6.3.3 **Decision point**: prototype sentuh data/payment? → trigger SOP-SK-03.
- 6.3.4 **Research note high-level WAJIB sebelum fitur dibangun** (kontrol SK-C7, GOVERNANCE §4.7) — kalau spike ini jadi dasar fitur yang bakal di-build (@killua/@saitama), arsipkan **research note** (evidence + opsi + rekomendasi, ikut poc-report-template) ke `senku/` + tautkan di `log.md`. **0 fitur boleh dibangun tanpa research note ter-arsip**.
- 6.3.5 **Exit criteria**: hipotesis ada verdict + evidence cited + time-spent vs estimate dicatat + recommendation actionable + (kalau jadi dasar fitur) research note ter-arsip + (kalau keep) hand-off plan jelas.

7. Pengecualian

- **Spike buntu sebelum time-box habis:** kalau jelas gak feasible lebih awal → STOP lebih cepat, report "refuted early" — gak buang sisa time-box.

8. Rekaman & Retensi

Rekaman	Lokasi	Retensi
Spike report (SPIKE-NNN)	log.md	Permanen
Research note high-level (sebelum fitur dibangun, §4.7)	senku/ + tautan log.md	Permanen
Throwaway code	team/senku/spikes/<date>-<topic>/	Sampai keputusan keep/discard
Hand-off plan (kalau keep)	log.md + mention @killua/@saitama	Permanen

9. Indikator Kinerja (KPI)

KPI	Rumus	Target
On-time-box	# spike selesai $\leq$ time-box $\div$ total	↑ tiap periode
Decisiveness	# spike kelar dengan verdict $\div$ total	100% (no ngambang)
Throwaway discipline	# prototype Senku ke-ship langsung ke prod	0

10. Riwayat Revisi

Rev	Tanggal	Perubahan
1.0	2026-05-29	Terbit pertama